

(*) On every Riemannian manifold, we will always use the Levi-Civita connection if there is no further comment.

1 The Curvature Tensor

Recall: For a finite-dimensional vector space V , there is a natural isomorphism between $T^{(k+1,l)}V$ and the space of multilinear maps

$$V^* \times \cdots \times V^* \times V \times \cdots \times V \rightarrow V$$

where V^* takes k copies and V takes l copies.

Recall: Recall that for a Riemannian manifold (M, g) , we defined the map

$$R : \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$$

$$(X, Y, Z) \mapsto \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$$

which is $C^\infty(M)$ -trilinear and thus defines a $(1, 3)$ -tensor field on M . The tensor field R is called the **(Riemann) curvature endomorphism** or **(1,3)-curvature tensor**. (We adopt the convention that the *last* index is the covariant upper one, which is contrary to our default assumption that covector arguments come first.) In terms of local coordinates (x^i) , the curvature endomorphism can be written as

$$R \triangleq R_{ijk}^l dx^i \otimes dx^j \otimes dx^k \otimes \frac{\partial}{\partial x^l}$$

i.e.,

$$R\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) \frac{\partial}{\partial x^k} = R_{ijk}^l \frac{\partial}{\partial x^l}$$

By calculation, we can get the formula for the Riemann curvature endomorphism in local coordinates

$$R_{ijk}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{jk}^m \Gamma_{im}^l - \Gamma_{ik}^m \Gamma_{jm}^l.$$

where we denote $\partial_i := \frac{\partial}{\partial x^i}$.

In fact, the curvature endomorphism also measures the failure of second covariants along along families of curves to commute.

Proposition 1.1. : Suppose (M, g) is a smooth Riemannian manifold, $\Gamma : J \times I \rightarrow M$ is a smooth one-parameter family of curves in M . Then for every smooth vector field V along Γ , we have

$$D_s D_t V - D_t D_s V = R(\partial_s \Gamma, \partial_t \Gamma) V.$$

Proof: For any $(s, t) \in J \times I$, choose a local coordinate (x^i) defined on a neighborhood of $\Gamma(s, t) \in M$ and write

$$\Gamma(s, t) \triangleq (\gamma^1(s, t), \dots, \gamma^n(s, t)), \quad V(s, t) \triangleq V^j(s, t) \partial_j|_{\Gamma(s, t)}$$

By the definition of covariant derivative along curves, we get

$$D_t V = \frac{\partial V^i}{\partial t} \partial_i + V^i D_t \partial_i$$

again, we get

$$D_s D_t V = \frac{\partial^2 V^i}{\partial s \partial t} \partial_i + \frac{\partial V^i}{\partial t} D_s \partial_i + \frac{\partial V^i}{\partial s} D_t \partial_i + V^i D_s D_t \partial_i.$$

Now, interchange the index s and t and subtract them, we get

$$D_s D_t V - D_t D_s V = V^i (D_s D_t \partial_i - D_t D_s \partial_i).$$

Denote

$$S := \partial_s \Gamma = \frac{\partial \gamma^k}{\partial s} \partial_k, \quad T := \partial_t \Gamma = \frac{\partial \gamma^j}{\partial t} \partial_j.$$

Since ∂_i is extendible, we have

$$D_t \partial_i = \nabla_T \partial_i = \frac{\partial \gamma^j}{\partial t} \nabla_{\partial_i} \partial_j$$

therefore, since $\nabla_{\partial_i} \partial_j$ is also extendible, we get

$$D_s D_t \partial_i = D_s \left(\frac{\partial \gamma^j}{\partial t} \nabla_{\partial_j} \partial_i \right) = \frac{\partial^2 \gamma^j}{\partial s \partial t} \nabla_{\partial_j} \partial_i + \frac{\partial \gamma^j}{\partial t} \nabla_s (\nabla_{\partial_i} \partial_j) = \frac{\partial^2 \gamma^j}{\partial s \partial t} \nabla_{\partial_j} \partial_i + \frac{\partial \gamma^j}{\partial t} \frac{\partial \gamma^k}{\partial s} \nabla_{\partial_k} \nabla_{\partial_i} \partial_j.$$

Now, interchange the index $s \leftrightarrow t$ and $j \leftrightarrow k$ and subtracting, we can get

$$D_s D_t \partial_i - D_t D_s \partial_i = \frac{\partial \gamma^j}{\partial t} \frac{\partial \gamma^k}{\partial s} (\nabla_{\partial_k} \nabla_{\partial_j} \partial_i - \nabla_{\partial_j} \nabla_{\partial_k} \partial_i) = \frac{\partial \gamma^j}{\partial t} \frac{\partial \gamma^k}{\partial s} R(\partial_k, \partial_j) \partial_i = R(S, T) \partial_i.$$

Hence we get the proposition. $\square$

Riemann Curvature Tensor

Definition 1.1. : We define the **Riemann curvature tensor** to be the $(0, 4)$ -tensor field Rm on the Riemannian manifold (M, g) , which is given by

$$Rm(X, Y, Z, W) := \langle R(X, Y)Z, W \rangle_g$$

for any $X, Y, Z, W \in \mathfrak{X}(M)$.

Remark: Actually, Rm is defined by $Rm := R^\flat$, which is obtained from the $(1, 3)$ -curvature tensor R by lowering its last index (recall that for a (k, l) -tensor F on M , we have the $(k-1, l+1)$ -tensor $F^\flat$ on M). So if we write Rm in local coordinates as

$$Rm \triangleq R_{ijkl} dx^i \otimes dx^j \otimes dx^k \otimes dx^l$$

then we have $R_{ijkl} = g_{lm} R_{ijk}^m$, hence we have

$$R_{ijkl} = g_{lm} (\partial_i \Gamma_{jk}^m - \partial_j \Gamma_{ik}^m + \Gamma_{jk}^p \Gamma_{ip}^m - \Gamma_{ik}^p \Gamma_{jp}^m).$$

2 Flat Manifolds

We say that a Riemannian manifold (M, g) is **flat** if it is locally isometric to a Euclidean space.

We say that a connection ∇ on a smooth manifold M satisfies the **flatness criterion** if for any $X, Y, Z \in \mathfrak{X}(M)$, we have the equality

$$\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z = \nabla_{[X, Y]} Z.$$

Recall: For a vector or tensor field V on the Riemannian manifold (M, g) , we say that V is **parallel** (w.r.t. the connection ∇) if it is parallel along every smooth curve in M .

Lemma 2.1. : *Suppose M is a smooth manifold, and ∇ is any connection on M satisfying the flatness criterion. Given $p \in M$ and any vector $v \in T_p M$, there exists a parallel vector field V on a neighborhood of p such that $V_p = v$.*

Proof: Let $p \in M$ and $v \in T_p M$, let (x^i) be any local coordinate on M centered at p . By shrinking the neighborhood if necessary, we may assume that the image of the coordinate map is an open cube $C_\varepsilon := \{x \mid |x^i| < \varepsilon, i = 1, 2, \dots, n\}$. We identify the coordinate domain with C_ε .

First, parallel transporting v along the x^1 -axis; then from each point on the x^1 -axis, parallel transport along the coordinate line parallel to the x^2 -axis; then successively parallel transport along the coordinate line parallel to the x^3 through x^n -axes. The result is a vector field V defined on in C_ε . By the theory from ODE about the smooth dependence of solutions to linear ODEs on initial conditions, we can know that V is smooth.

We know that $\nabla_X V$ is linear over $C^\infty(M)$ in X , so in order to show that V is parallel, we only need to show that $\nabla_{\partial_i} V = 0$ for each $i = 1, 2, \dots, n$. By the construction of V , we know that $\nabla_{\partial_1} V = 0$ on the x^1 -axis, $\nabla_{\partial_2} V = 0$ on the (x^1, x^2) -plane, in general $\nabla_{\partial_k} V = 0$ on the slice $M_k \subseteq C_\varepsilon$ defined by $x^{k+1} = x^{k+2} = \dots = x^n = 0$. Now we prove the following fact by induction on k : $\nabla_{\partial_1} V = \dots = \nabla_{\partial_k} V = 0$ on M_k . For $k = 1$, it's clear by the construction of V ; now we assume that this equality holds for some k . By construction of V , we know that $\nabla_{\partial_{k+1}} V = 0$ on all of M_{k+1} , and for $i \leq k$, the inductive hypothesis shows that $\nabla_{\partial_i} V = 0$ on the hyperplane $M_k \subseteq M_{k+1}$.

Since $[\partial_i, \partial_j] = 0$, the flatness criterion gives

$$\nabla_{\partial_{k+1}}(\nabla_{\partial_i} V) = \nabla_{\partial_i}(\nabla_{\partial_{k+1}} V) = 0 \quad \text{on } M_{k+1}.$$

this means that $\nabla_{\partial_i} V$ is parallel along the x^{k+1} -curves starting at M_k . Since $\nabla_{\partial_i} V$ vanishes on M_k and the zero vector field is the unique parallel transport of zero, we conclude that $\nabla_{\partial_i} V$ is zero on each x^{k+1} -curve. Since every point of M_{k+1} is on one of these curves, it follows that $\nabla_{\partial_i} V = 0$ on all of M_{k+1} . $\square$

Recall: Recall that the Euclidean connection $\bar{\nabla}$ on $\mathbb{R}^n$ is given by

$$\bar{\nabla}_X Y = X(Y^i) \frac{\partial}{\partial x^i}$$

for any vector fields X and $Y = Y^i \frac{\partial}{\partial x^i}$. Thus we can see that

$$\bar{\nabla}_X \bar{\nabla}_Y Z - \bar{\nabla}_Y \bar{\nabla}_X Z = (XY(Z^k) - YX(Z^k)) \frac{\partial}{\partial x^k} = \bar{\nabla}_{[X,Y]} Z.$$

which means that the Euclidean connection satisfies the flatness criterion. Moreover, we have known that the Euclidean connection on $\mathbb{R}^n$ is precisely the Levi-Civita connection on it.

Recall: (**Naturality of Levi-Civita Connection**) Suppose (M, g) and $(\tilde{M}, \tilde{g})$ are Riemannian manifolds, let ∇ be the Levi-Civita connection of g and $\tilde{\nabla}$ be that of $\tilde{g}$, if $\varphi : M \rightarrow \tilde{M}$ is an isometry, then we have $\varphi^* \tilde{\nabla} = \nabla$.

Theorem 2.1. : *A Riemannian manifold is flat $\iff$ its curvature tensor vanishes identically.*

Proof: ($\Rightarrow$): Since the Euclidean metric on $\mathbb{R}^n$ satisfies the flatness criterion, by the naturality, the Levi-Civita connection on every manifold that is locally isometric to a Euclidean space must also satisfy the same equation, so its curvature endomorphism must be zero and hence so is its curvature tensor.

($\Leftarrow$): Now suppose (M, g) has zero curvature tensor, which means that its curvature endomorphism vanishes as well, thus the Levi-Civita connection satisfies the flatness criterion. We first show that g admits a parallel orthonormal frame in a neighborhood of each point.

Take $\forall p \in M$, choose an orthonormal basis $(b_1, \dots, b_n)$ for $T_p M$. By Lemma 2.1, there exists parallel vector fields $E_1, \dots, E_n$ on a neighborhood of p such that $E_i|_p = b_i$ for $i = 1, 2, \dots, n$. Since parallel transport preserves the inner products, the vector fields (E_i) are orthonormal in all of U . We know that the Levi-Civita connection is symmetric, so we have

$$[E_i, E_j] = \nabla_{E_i} E_j - \nabla_{E_j} E_i = 0.$$

hence the vector fields $(E_1, \dots, E_n)$ form a commuting orthonormal frame on U , by the knowledge of smooth manifolds, there exists a coordinate $(y^1, y^2, \dots, y^n)$ on a neighborhood of p such that $E_i = \frac{\partial}{\partial y^i}$. In any such coordinates, $g_{ij} = g(\partial_i, \partial_j) = g(E_i, E_j) = \delta_{ij}$. So the map $y = (y^1, \dots, y^n)$ is an isometry from a neighborhood of p to an open subset of the Euclidean space. $\square$

3 Symmetries of the Curvature Tensor

Symmetries of the Curvature Tensor

Proposition 3.1. : *Let (M, g) be a Riemannian manifold, the $(0, 4)$ -curvature tensor of g has the following symmetries for all vector fields W, X, Y, Z :*

- (1) $Rm(W, X, Y, Z) = -Rm(X, W, Y, Z)$
- (2) $Rm(W, X, Y, Z) = -Rm(W, X, Z, Y)$
- (3) $Rm(W, X, Y, Z) = Rm(Y, Z, W, X)$
- (4) $Rm(W, X, Y, Z) + Rm(X, Y, W, Z) + Rm(Y, W, X, Z) = 0$ (algebraic Bianchi Identity)

Proof: (1) By the definition of Rm and the property of R ;

(2) By the compatibility of the Levi-Civita connection with the metric, we get

$$WX|Y|^2 = W(\nabla_X \langle Y, Y \rangle) = W(2\langle \nabla_X Y, Y \rangle) = 2\langle \nabla_W \nabla_X Y, Y \rangle + 2\langle \nabla_X Y, \nabla_W Y \rangle.$$

$$XW|Y|^2 = X(\nabla_W \langle Y, Y \rangle) = X(2\langle \nabla_W Y, Y \rangle) = 2\langle \nabla_X \nabla_W Y, Y \rangle + 2\langle \nabla_W Y, \nabla_X Y \rangle.$$

$$[W, X]|Y|^2 = \nabla_{[W, X]} \langle Y, Y \rangle = 2\langle \nabla_{[W, X]} Y, Y \rangle.$$

We subtract the second and third equations from the first, we get

$$0 = 2\langle \nabla_W \nabla_X Y, Y \rangle - 2\langle \nabla_X \nabla_W Y, Y \rangle - 2\langle \nabla_{[W, X]} Y, Y \rangle = 2\langle R(W, X)Y, Y \rangle = 2Rm(W, X, Y, Y).$$

Since the vector field Y is arbitrary, so we also have

$$Rm(W, X, Y + Z, Y + Z) = 0$$

which implies that $Rm(W, X, Y, Z) = -Rm(W, X, Z, Y)$;

(4) By the definition of R , we have

$$\begin{aligned} R(W, X)Y + R(X, Y)W + R(Y, W)X &= (\nabla_W \nabla_X Y - \nabla_X \nabla_W Y - \nabla_{[W, X]} Y) + (\nabla_X \nabla_Y W - \nabla_Y \nabla_X W - \nabla_{[X, Y]} W) \\ &+ (\nabla_Y \nabla_W X - \nabla_W \nabla_Y X - \nabla_{[Y, W]} X) = \nabla_W (\nabla_X Y - \nabla_Y X) + \nabla_X (\nabla_Y W - \nabla_W Y) + \nabla_Y (\nabla_W X - \nabla_X W) \\ &- \nabla_{[W, X]} Y - \nabla_{[X, Y]} W - \nabla_{[Y, W]} X = \nabla_W [X, Y] + \nabla_X [Y, W] + \nabla_Y [W, X] - \nabla_{[W, X]} Y - \nabla_{[X, Y]} W - \nabla_{[Y, W]} X \\ &= [W, [X, Y]] + [X, [Y, W]] + [Y, [W, X]] = 0 \end{aligned}$$

where the last equation is just the Jacobi identity. Thus we have

$$\langle R(W, X)Y + R(X, Y)W + R(Y, W)X, Z \rangle = Rm(W, X, Y, Z) + Rm(X, Y, W, Z) + Rm(Y, W, X, Z) = 0.$$

(3) By (4), we have the followign equations

$$Rm(W, X, Y, Z) + Rm(X, Y, W, Z) + Rm(Y, W, X, Z) = 0$$

$$Rm(X, Y, Z, W) + Rm(Y, Z, X, W) + Rm(Z, X, Y, W) = 0$$

$$Rm(Y, Z, W, X) + Rm(Z, W, Y, X) + Rm(W, Y, Z, X) = 0$$

$$Rm(Z, W, X, Y) + Rm(W, X, Z, Y) + Rm(X, Z, W, Y) = 0$$

Add up all the four equations and apply (2) four times makes all the terms in the first two columns cancel. Then apply (1) and (2) in the last column yields $2Rm(Y, W, X, Z) - 2Rm(X, Z, Y, W) = 0$, which is equal to (3). $\square$

Remark: It is useful to record the form of these symmetries in terms of components with respect to any basis:

$$(a) R_{ijkl} = -R_{jikl}$$

$$(b) R_{ijkl} = -R_{ijlk}$$

$$(c) R_{ijkl} = R_{klij}$$

$$(d) R_{ijkl} + R_{jkil} + R_{kijl} = 0$$

Differential Bianchi Identity

Proposition 3.2. : *The total covariant derivative of the curvature tensor satisfies the following identity:*

$$\nabla Rm(X, Y, Z, V, W) + \nabla Rm(X, Y, V, W, Z) + \nabla Rm(X, Y, W, Z, V) = 0.$$

Proof: Let p an arbitrary point in the Riemannian manifold M , let (x^i) be normal coordinates centered at p and let X, Y, Z, V, W be arbitrary coordinate basis vector fields, so by the properties of normal coordinates, we know that $[\partial_i, \partial_j] \equiv 0$ and $\Gamma_{ij}^k(p) = 0$.

Now, by the symmetries of the curvature tensor, consider the equation

$$\begin{aligned} & \nabla Rm(X, Y, Z, V, W) + \nabla Rm(X, Y, V, W, Z) + \nabla Rm(X, Y, W, Z, V) \\ &= \nabla Rm(Z, V, X, Y, W) + \nabla Rm(V, W, X, Y, Z) + \nabla Rm(W, Z, X, Y, V) \end{aligned}$$

Use the compatibility with the metric and the observations above, the first term in the right-hand side evaluated at p is

$$\begin{aligned} (\nabla_W Rm)(Z, V, X, Y) &= W(Rm(Z, V, X, Y)) = \nabla_W(Rm(Z, V, X, Y)) = \nabla_W \langle R(Z, V)X, Y \rangle \\ &= \nabla_W \langle \nabla_Z \nabla_V X - \nabla_V \nabla_Z X - \nabla_{[Z, V]} X, Y \rangle = \langle \nabla_W \nabla_Z \nabla_V X - \nabla_W \nabla_V \nabla_Z X, Y \rangle. \end{aligned}$$

Write this equation three times with the vector fields W, Z, V cyclically permuted and sum all three gives

$$\begin{aligned} & \nabla Rm(Z, V, X, Y, W) + \nabla Rm(V, W, X, Y, Z) + \nabla Rm(W, Z, X, Y, V) \\ &= \langle \nabla_W \nabla_Z \nabla_V X - \nabla_W \nabla_V \nabla_Z X + \nabla_Z \nabla_V \nabla_W X - \nabla_Z \nabla_W \nabla_V X + \nabla_V \nabla_W \nabla_Z X - \nabla_V \nabla_Z \nabla_W X, Y \rangle \\ &= \langle R(W, Z)(\nabla_V X) + R(Z, V)(\nabla_W X) + R(V, W)(\nabla_Z X), Y \rangle = 0 \end{aligned}$$

the last equality holds because by our assumption we have $\nabla_V X = \nabla_W X = \nabla_Z X = 0$. $\square$

Remark: In components, this proposition is

$$R_{ijkl;m} + R_{ijlm;k} + R_{ijmk;l} = 0.$$

参考文献

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